THE STRONG COMPLETION TOWER: BAR-COBAR DUALITY FOR NON-QUADRATIC CHIRAL ALGEBRAS

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Abstract. The classical bar-cobar adjunction of Priddy produces a Koszul dual for a quadratic algebra. For non-quadratic chiral algebras on a curve, the bar complex is infinite-dimensional in every bar length and the naive adjunction fails to converge. We introduce the *strong completion tower*: a filtered augmented curved chiral A_∞-algebra whose operations are filtration-nondecreasing on the augmentation ideal. Under this hypothesis we prove a degree cutoff lemma showing that each graded component of the Maurer-Cartan equation involves only finitely many terms, that Mittag-Leffler holds automatically, and that bar-cobar inversion extends to the completion as a quasi-isomorphism of curved dg chiral coalgebras and algebras. The result upgrades the previously conjectural MC4 completion closure to a theorem with sharp hypotheses and splits the completion problem into a positive regime ($\mathrm{MC4}^+$), unconditionally solved, and a weight-zero resonant regime ($\mathrm{MC4}^0$), tractable for finite resonance rank. The W_N tower with principal finite-type stages is rigid, resolving twenty-one standing conjectures. Virasoro, affine Kac-Moody at non-critical level, principal $\mathcal{W}$-algebras, and lattice vertex algebras are strong completion towers.

1. Introduction

1.1. **The quadratic case and its limits.** Classical Koszul duality is a phenomenon of graded algebras. A quadratic algebra $A = T(V)/(R)$ determines a dual coalgebra $A^{!`} = T^c(sV)/(sR)$; the bar construction $B(A)$ mediates between them; and $A$ is Koszul when $H^*(B(A))$ concentrates in bar length one, so that $\Omega(A^{!`}) \simeq A$. The theory is due to Priddy [5], Beilinson-Ginzburg-Soergel, and in operadic form Loday-Vallette [2].

On an algebraic curve X, the objects of interest are chiral (vertex) algebras, whose relations are never quadratic in the strict sense. Heisenberg has a central term in the $J \cdot J$ OPE. Virasoro has a quartic pole. Principal $\mathcal{W}$-algebras $\mathcal{W}^k(\mathfrak{g}, f_{\mathrm{prin}})$ have OPE poles of order up to $2\,\mathrm{rk}(\mathfrak{g}) + 2$. Affine Yangians and quantum toroidal algebras have spectral-parameter relations of unbounded polynomial order. In each case the bar complex is a perfectly good

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dg coalgebra, but the iterated OPE produces bar elements whose differential contains infinitely many summands. No finite-dimensional truncation sees the whole picture, and the classical bar-cobar adjunction is silent.

1.2. **The completion problem.** The remedy is completion. One introduces a decreasing filtration $\mathcal{A} = F^0\mathcal{A} \supset F^1\mathcal{A} \supset \cdots$ with finite-dimensional quotients $\mathcal{A}_{\leq N} := \mathcal{A}/F^{N+1}\mathcal{A}$ and writes $\widehat{\overline{B}}(\mathcal{A}) := \varprojlim_N \overline{B}(\mathcal{A}_{\leq N})$. At each finite stage N the strict bar-cobar adjunction produces a universal twisting morphism $\tau_N \colon \overline{B}(\mathcal{A}_{\leq N}) \to \mathcal{A}_{\leq N}$ which is a Maurer-Cartan element of the convolution dg Lie algebra $\mathrm{Conv}(\overline{B}(\mathcal{A}_{\leq N}), \mathcal{A}_{\leq N})$. The completion problem, labelled MC4 in [3], asks: do the finite-stage MC elements τ_N glue to a single MC element $\widehat{\tau}$ in the completed convolution algebra, and is the twisted tensor product acyclic in the limit?

Two obstructions appear. First, the MC equation $\partial(\widehat{\tau}) + \widehat{\tau} \star \widehat{\tau} = 0$ is a sum over all bar degrees, and a priori each component is infinite. Second, the passage from finite-stage acyclicity to acyclicity in the limit requires the Mittag-Leffler condition, which is not automatic.

1.3. **The main result.** The central observation of this paper is that a single axiom on the filtration makes both obstructions evaporate. The axiom is that the A_∞ operations of $\mathcal{A}$ respect the additive weight:

$$\mu_r(F^{i_1}\mathcal{A}, \ldots, F^{i_r}\mathcal{A}) \subset F^{i_1+\cdots+i_r}\mathcal{A}. \tag{1.1}$$

A filtered curved chiral A_∞-algebra for which (1.1) holds and whose augmentation ideal $\overline{\mathcal{A}} = F^1\mathcal{A}$ is the first filtration piece is a *strong completion tower* (Definition 3.1).

Theorem 1.1 (Main theorem, informal statement)**.** *Let $\mathcal{A}$ be a strong completion tower whose finite-stage quotients lie in the proved bar-cobar regime. Then the completed bar construction $\widehat{\overline{B}}(\mathcal{A})$ is a separated complete pronilpotent curved dg chiral coalgebra, the completed convolution algebra carries a canonical Maurer-Cartan element $\widehat{\tau} = \varprojlim_N \tau_N$, Mittag-Leffler holds automatically, and the counit*

$$\widehat{\epsilon} \colon \widehat{\Omega}\big(\widehat{\overline{B}}(\mathcal{A})\big) \xrightarrow{\sim} \mathcal{A}$$

is a quasi-isomorphism. Completed bar and completed cobar are quasi-inverse equivalences on the full subcategory of strong completion towers.

The precise statement is Theorem 6.1 below. The proof assembles three ingredients: (i) the degree cutoff lemma (§4), which reduces each component of the MC equation to a finite sum; (ii) the automatic Mittag-Leffler property (§5), a direct consequence of the strong filtration axiom; and (iii) the reduction principle of Lemma 5.2 which turns the finite-stage theory into a limit statement via the Milnor exact sequence.

1.4. **Positive versus resonant towers.** The strong filtration axiom naturally partitions completion towers into two regimes. A tower is *positive* if the filtration is defined by a genuine positive grading (e.g. the L_0-eigenvalue filtration on a graded vertex algebra). The degree cutoff lemma then holds trivially and the completion is unconditionally solved. We call this the *MC4*$^+$ *problem*: it is now closed.

A tower is *resonant* if there is a finite-dimensional weight-zero piece $R_{\mathcal{A}}$ on which higher operations can preserve filtration degree. The *resonance rank*

$$\rho(\mathcal{A}) := \dim H^*(R_{\mathcal{A}}, m_1^{R_{\mathcal{A}}})$$

classifies the completion difficulty. Positive towers have $\rho = 0$; Virasoro and the principal $\mathcal{W}_N$-algebras have $0 < \rho < \infty$; genuinely wild algebras have $\rho = \infty$. The MC4^0 problem concerns the middle regime. For the standard principal finite-type W_N stages, the resonance is rigid: the strong filtration axiom plus weight-cutoff stabilization force the completion data. This W_N rigidity resolves twenty-one standing conjectures in the completion frontier, catalogued in §8.

1.5. **Relation to prior work and organization.** Positselski [4] developed curved bar constructions for associative algebras with curvature. Loday-Vallette [2] supply the operadic bar-cobar machinery on which the chiral lift is built. The Beilinson-Drinfeld framework of filtered chiral algebras with I-adic topology [1] provides the ambient topology for the inverse limit. The present theorem is the chiral and A_∞ lift: the novelty is the strong filtration axiom, which identifies the precise structural hypothesis under which all finite-stage data assemble without further input.

Section 2 fixes notation. Section 3 defines the strong completion tower. Section 4 proves the degree cutoff lemma. Section 5 proves Mittag-Leffler. Section 6 assembles the main theorem. Section 7 discusses the MC4$^+$ versus MC4^0 dichotomy and the W_N rigidity. Section 8 works out the standard families.

2. The bar construction on the augmentation ideal

Throughout, X is a smooth complex algebraic curve and $\mathcal{A}$ is an augmented curved chiral A_∞-algebra on X in the sense of the monograph [3]. Augmentation means that $\mathcal{A}$ is equipped with a counit $\varepsilon\colon \mathcal{A} \to \mathbf{1}$ splitting the unit. The augmentation ideal is

$$\overline{\mathcal{A}} := \ker(\varepsilon).$$

Remark 2.1 (Augmentation ideal). The bar construction is built on $\overline{\mathcal{A}}$, not on $\mathcal{A}$: writing $T^c(s^{-1}\mathcal{A})$ in place of $T^c(s^{-1}\overline{\mathcal{A}})$ includes the unit and breaks the Koszul sign computation. We therefore set

$$\overline{B}^{\mathrm{ch}}(\mathcal{A}) := T^c\big(s^{-1}\overline{\mathcal{A}}\big),$$

with s^{-1} the homological desuspension $|s^{-1}v| = |v| - 1$ and with the chiral coproduct obtained by deconcatenation on the collision divisor of $\overline{C}_{\bullet+1}(X)$.

The curved chiral A_∞ operations $\{\mu_r\}_{r\geq 0}$ assemble into a codifferential $d_{\overline{B}}$ on $\overline{B}^{\mathrm{ch}}(\mathcal{A})$ with curvature coming from μ_0, so $\overline{B}^{\mathrm{ch}}(\mathcal{A})$ is a curved dg coalgebra. The convolution dg Lie algebra $\mathrm{Conv}(\mathcal{C}, \mathcal{A}) := \mathrm{Hom}(\mathcal{C}, \mathcal{A})$ has bracket from the coproduct on $\mathcal{C}$ and the product on $\mathcal{A}$; a Maurer-Cartan element is a degree-1 map $\tau\colon \mathcal{C} \to \mathcal{A}$ with $\partial(\tau) + \tau \star \tau = 0$.

Theorem 2.2 (Finite-type bar-cobar duality, cited)**.** *Let $\mathcal{A}$ be a finite-type augmented curved chiral A_∞-algebra on X in the proved bar-cobar regime. Then the universal twisting morphism $\tau\colon \overline{B}^{\mathrm{ch}}(\mathcal{A}) \to \mathcal{A}$ is a Maurer-Cartan element of* $\mathrm{Conv}(\overline{B}^{\mathrm{ch}}(\mathcal{A}), \mathcal{A})$*, and the counit*

$$\epsilon\colon \Omega^{\mathrm{ch}}(\overline{B}^{\mathrm{ch}}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$$

is a quasi-isomorphism of curved dg chiral algebras.

Proof. See [3, Thm. B]. The finite-type hypothesis reduces to finite-type vertex algebras, where the operadic bar-cobar machinery of Loday-Vallette [2] applies after the curved extension of Positselski [4]. □

3. The strong filtration axiom

Definition 3.1 (Strong completion tower)**.** An augmented curved chiral A_∞-algebra $\mathcal{A}$ on a curve X is a *strong completion tower* if it carries a descending filtration

$$\mathcal{A} = F^0\mathcal{A} \supset F^1\mathcal{A} \supset F^2\mathcal{A} \supset \cdots, \qquad \bigcap_{N\geq 0} F^{N+1}\mathcal{A} = 0,$$

such that:

(1) $\mathcal{A}$ is separated and complete: $\mathcal{A} \cong \varprojlim_N \mathcal{A}_{\leq N}$, where $\mathcal{A}_{\leq N} := \mathcal{A}/F^{N+1}\mathcal{A}$;
(2) every quotient $\mathcal{A}_{\leq N}$ is of finite type and lies in the proved bar-cobar regime of Theorem 2.2;
(3) $\overline{\mathcal{A}} = F^1\mathcal{A}$: the augmentation ideal is the first filtration piece;
(4) all chiral A_∞-operations are filtration-nondecreasing,

$$\mu_r(F^{i_1}\mathcal{A}, \ldots, F^{i_r}\mathcal{A}) \subset F^{i_1+\cdots+i_r}\mathcal{A}. \tag{3.1}$$

Axiom (3) ties the filtration to the augmentation. Axiom (4) is the structural heart of the definition: it says that the higher operations respect the additive grading.

Remark 3.2 (On the quotient maps)*.* Each projection $p_N\colon \mathcal{A}_{\leq N+1} \twoheadrightarrow \mathcal{A}_{\leq N}$ is automatically a strict morphism of curved chiral A_∞-algebras, because the kernel $F^{N+1}\mathcal{A}/F^{N+2}\mathcal{A}$ is an A_∞-ideal of $\mathcal{A}_{\leq N+1}$ by (3.1). The finite-stage bar constructions $\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N})$ therefore form an inverse system of curved dg chiral coalgebras with strict morphisms, and the completed bar construction is the literal inverse limit.

Proposition 3.3 (Standard landscape examples)**.** *The following are strong completion towers with respect to the conformal-weight filtration* $F^N\mathcal{A} := \bigoplus_{h\geq N}\mathcal{A}_h$*:*

(a) *the affine Kac-Moody vertex algebra* $V_k(\mathfrak{g})$ *for any simple* $\mathfrak{g}$ *and any non-critical level* $k \neq -h^\vee$*;*
(b) *the Virasoro vertex algebra* Vir_c *for any* c*;*
(c) *the principal* $\mathcal{W}$*-algebra* $\mathcal{W}^k(\mathfrak{g}, f_{\mathrm{prin}})$ *for any simple* $\mathfrak{g}$ *and any non-critical* k*;*
(d) *the lattice vertex algebra* V_Λ *for any positive-definite even lattice* Λ*.*

Proof. Each family admits a $\mathbb{Z}_{\geq 0}$-grading by L_0-eigenvalue with finite-dimensional graded pieces and vacuum at $h = 0$, so axioms (1)–(3) hold at once. For axiom (4), the bar differential extracts the coefficient of $(z-w)^{-1}$ in the OPE after desuspension, and this residue operation preserves the total weight on the augmentation ideal $\overline{\mathcal{A}} = F^1\mathcal{A}$ where all inputs have strictly positive weight. Iterating propagates the bound to $\mu_r(F^{i_1}, \dots, F^{i_r}) \subset F^{i_1+\cdots+i_r}$.

Family-by-family: affine Kac-Moody has weight-1 generators J^a with OPE poles of order ≤ 2; Virasoro has a weight-2 generator L with poles of order ≤ 4; principal $\mathcal{W}^k(\mathfrak{g}, f_{\mathrm{prin}})$ has weight-s generators for $s = 2, \dots, \mathrm{rk}(\mathfrak{g}) + 1$ with poles of order $\leq 2s$; and V_Λ has weight-1 Cartan generators together with vertex operators e^α of weight $|\alpha|^2/2 \geq 1$. □

Example 3.4 (Non-example: $\beta\gamma$-system with conformal grading)**.** The $\beta\gamma$-system has a weight-$\frac{1}{2}$ grading that refines the conformal grading. With respect to the conformal-weight filtration alone, the strong filtration axiom fails because the OPE $\beta(z)\gamma(w) \sim 1/(z-w)$ produces a weight-zero output from weight-$\frac{1}{2}$ inputs, violating additivity. The $\beta\gamma$-system is instead a strong completion tower with respect to the fermion-number filtration.

4. The degree cutoff lemma

The structural consequence of the strong filtration axiom is that in the finite-stage bar complex, only finitely many degrees contribute. This is the precise mechanism by which the MC equation becomes componentwise finite.

Lemma 4.1 (Degree cutoff)**.** *Let* $\mathcal{A}$ *be a strong completion tower and let* $\mathcal{A}_{\leq N} = \mathcal{A}/F^{N+1}\mathcal{A}$ *be the* N*-th quotient. In the finite-stage convolution algebra* $\mathrm{Conv}(\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N}), \mathcal{A}_{\leq N})$*, the Maurer-Cartan equation*

$$\partial(\tau_N) + \tau_N \star \tau_N = 0$$

involves only degrees $r \leq N$*: on* $\mathcal{A}_{\leq N}$ *the bar differential is the finite sum* $b_1 + b_2 + \cdots + b_N$ *of coderivations coming from* $\mu_1, \dots, \mu_N$*.*

Proof. Every bar input lies in $\overline{\mathcal{A}}_{\leq N}$, which by axiom (3) equals the image of $F^1\mathcal{A}$ in the quotient. The strong filtration axiom (3.1) gives

$$\mu_r(\overline{\mathcal{A}}^{\otimes r}) \subset \mu_r\big((F^1\mathcal{A})^{\otimes r}\big) \subset F^r\mathcal{A}.$$

Reducing modulo $F^{N+1}\mathcal{A}$, any term with $r \geq N+1$ lands in $F^r\mathcal{A} \subset F^{N+1}\mathcal{A}$ and vanishes in $\mathcal{A}_{\leq N}$. So the induced bar differential on $\mathcal{A}_{\leq N}$ is the finite sum $b_1 + b_2 + \cdots + b_N$, as claimed. □

Remark 4.2 (Degree cutoff as PBW degeneration). The degree cutoff is the operadic counterpart of the classical PBW theorem. For the universal enveloping algebra $U(\mathfrak{g})$ with its standard filtration by number of generators, the associated graded is $\mathrm{Sym}(\mathfrak{g})$; higher-degree brackets vanish modulo lower filtration stages. The strong filtration axiom extends this phenomenon to all chiral A_∞-operations on a graded vertex algebra: the degree cutoff lemma is the statement that at each finite stage, only finitely many Poisson (or vertex) operations are active. The classical PBW theorem is the $r = 2$ case.

Corollary 4.3 (Finite-stage MC assembly)**.** *Let $\mathcal{A}$ be a strong completion tower. The universal twisting morphism τ_N at stage N is a Maurer-Cartan element of a* finite-dimensional *convolution A_∞-algebra: the quotient $\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N})$ has finite-dimensional bar-length components, and the bar differential is the finite sum of Lemma 4.1.*

Proof. Finite type of $\mathcal{A}_{\leq N}$ gives finite dimension of each $\overline{\mathcal{A}}_{\leq N}^{\otimes r}$ for every r, and the degree cutoff lemma bounds the number of nonzero bar differential terms. Theorem 2.2 then gives the Maurer-Cartan property. □

5. Mittag-Leffler is automatic

The second obstruction to passing from finite-stage acyclicity to acyclicity in the limit is the possible failure of the Mittag-Leffler condition on the inverse system of finite-stage cohomologies. We show that the strong filtration axiom makes it automatic.

Proposition 5.1 (Mittag-Leffler for strong towers)**.** *Let $\mathcal{A}$ be a strong completion tower. For every cohomological degree m, the inverse system*

$$\left\{H^m(\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N}))\right\}_{N\geq 0}$$

satisfies the Mittag-Leffler condition.

Proof. The transition maps $\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N+1}) \twoheadrightarrow \overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N})$ are surjections, because the finite-stage bar construction is functorial on surjections of finite-type curved chiral A_∞-algebras and $p_N \colon \mathcal{A}_{\leq N+1} \twoheadrightarrow \mathcal{A}_{\leq N}$ is itself surjective.

The key observation is that the strong filtration axiom produces a splitting of the inverse system by weight. For each fixed conformal weight w, the weight-w subcomplex of $\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N})$ involves only bar entries of total weight w, and such entries can only use generators of weight $\leq w$. Once $N \geq w$, all such generators already occur in $\mathcal{A}_{\leq N}$, so the transition map

$$\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N+1})_w \xrightarrow{\sim} \overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N})_w$$

is an isomorphism of subcomplexes. Passing to cohomology gives stable isomorphisms for $N \geq w$, hence the weight-w cohomology slice stabilizes at $N = w$.

Summing over weights, the full cohomology $H^m(\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N}))$ is a direct sum of finite-dimensional weight slices each of which stabilizes at a finite stage. In particular the transition maps are eventually isomorphisms on every fixed weight slice, so the inverse system is Mittag-Leffler: the derived limit $\varprojlim_N^1 H^m$ vanishes. □

Lemma 5.2 (Complete filtered comparison)**.** *Let $\{f_N \colon C_N \to D_N\}$ be a morphism of inverse systems of complexes with surjective transition maps. If each f_N is a quasi-isomorphism and the inverse systems $\{H^*(C_N)\}$ and $\{H^*(D_N)\}$ are Mittag-Leffler, then the induced map on limits $\widehat{f} \colon \widehat{C} \to \widehat{D}$ is a quasi-isomorphism.*

Proof. For any inverse system of cochain complexes the Milnor exact sequence reads

$$0 \to \varprojlim_N^1 H^{m-1}(C_N) \to H^m(\widehat{C}) \to \varprojlim_N H^m(C_N) \to 0.$$

Mittag-Leffler kills the derived limit term, and the map on honest limits is an isomorphism because each f_N is. □

6. The main theorem

Theorem 6.1 (Strong completion tower bar-cobar duality)**.** *Let $\mathcal{A}$ be a strong completion tower (Definition 3.1). Write $\tau_N \in \mathrm{MC}\big(\mathrm{Conv}(\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N}), \mathcal{A}_{\leq N})\big)$ for the universal twisting morphism at stage N. Then:*

(1) Completed coalgebra. *The completed bar construction*

$$\widehat{\overline{B}}^{\mathrm{ch}}(\mathcal{A}) := \varprojlim_N \overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N})$$

exists as a separated complete pronilpotent curved dg chiral coalgebra with continuous differential, coproduct, and curvature. At each stage, $\widehat{\overline{B}}^{\mathrm{ch}}(\mathcal{A})/F^{N+1} \cong \overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N})$.

(2) Completed MC element. *The componentwise limit $\widehat{\tau} = \varprojlim_N \tau_N$ is a Maurer-Cartan element of the completed convolution algebra*

$$\widehat{\mathrm{Conv}} := \varprojlim_N \mathrm{Conv}(\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N}), \mathcal{A}_{\leq N}),$$

and the MC equation $\partial(\widehat{\tau}) + \widehat{\tau} \star \widehat{\tau} = 0$ converges in the inverse-limit topology, each component being a finite sum by the degree cutoff lemma.

(3) Completed cobar. *The completed cobar object*

$$\widehat{\Omega}^{\mathrm{ch}}\big(\widehat{\overline{B}}^{\mathrm{ch}}(\mathcal{A})\big) := \varprojlim_N \Omega^{\mathrm{ch}}(\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N}))$$

is a separated complete curved dg chiral algebra, equal to the twisted tensor product $\mathcal{A} \otimes_{\widehat{\tau}} \widehat{\overline{B}}^{\mathrm{ch}}(\mathcal{A})$.

(4) Counit is a quasi-isomorphism. *The completed counit*

$$\widehat{\epsilon}\colon \widehat{\Omega}^{\mathrm{ch}}\big(\widehat{\overline{B}}^{\mathrm{ch}}(\mathcal{A})\big) \xrightarrow{\sim} \mathcal{A}$$

is a quasi-isomorphism of curved dg chiral algebras.

(5) Dual quasi-isomorphism. *If* $\mathcal{C} = \varprojlim_N \mathcal{C}_{\leq N}$ *is a separated complete pronilpotent curved dg chiral coalgebra whose finite quotients lie in the theorematic regime, then the completed unit*

$$\widehat{\eta}\colon \mathcal{C} \xrightarrow{\sim} \widehat{\overline{B}}^{\mathrm{ch}}\big(\widehat{\Omega}^{\mathrm{ch}}(\mathcal{C})\big)$$

is a quasi-isomorphism.

(6) Quasi-inverse equivalences. *Completed bar and completed cobar are quasi-inverse equivalences on the full subcategory of strong completion towers whose finite quotients lie in the theorematic regime, exhibiting completed Koszul duality in the non-quadratic curved regime.*

Proof. Step 1: the completed coalgebra. Each quotient map $p_N\colon \mathcal{A}_{\leq N+1} \twoheadrightarrow \mathcal{A}_{\leq N}$ is strict by axiom (4), inducing a strict morphism $q_N\colon \overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N+1}) \to \overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N})$ compatible with differential, coproduct, coaugmentation, and curvature. The inverse limit exists componentwise with $b = \varprojlim b_N$, $\Delta = \varprojlim \Delta_N$, $h = \varprojlim h_N$; the curved coalgebra identities hold quotientwise and hence on the limit. Continuity is formal for the inverse-limit topology. Pronilpotence: the reduced coproduct on bar-length-$\leq m$ elements is nilpotent on each finite quotient, so the limit is topologically pronilpotent.

Step 2: MC assembly. Define $\widehat{\tau} := \varprojlim_N \tau_N$. By Lemma 4.1, modulo $F^{N+1}\mathcal{A}$ only degrees $\leq N$ contribute to $\widehat{\tau} \star \widehat{\tau}$. The MC equation holds modulo F^{N+1} (where it reduces to the finite-stage equation for τ_N), hence on the completed coalgebra. This is claim (2).

Step 3: completed cobar. Each $\Omega^{\mathrm{ch}}(\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N}))$ is a curved dg chiral algebra; compatible transition maps give a complete curved dg chiral algebra in the limit, with continuity automatic. This is claim (3).

Step 4: counit is a quasi-isomorphism. The quotient of $\widehat{\epsilon}$ modulo F^{N+1} is the finite-stage counit ϵ_N, which is a quasi-isomorphism by Theorem 2.2. By Proposition 5.1, the inverse system $\{H^m(\mathcal{A}_{\leq N})\}$ is Mittag-Leffler. Applying Lemma 5.2 to the morphism $\{\epsilon_N\}$ of inverse systems yields claim (4).

Step 5: dual quasi-isomorphism. The argument for $\widehat{\eta}$ is identical with bar and cobar exchanged, giving claim (5).

Step 6: quasi-inverse equivalences. Claims (4) and (5) are the two triangle identities up to quasi-isomorphism, so completed bar and completed cobar form an adjoint equivalence on the homotopy categories restricted to strong completion towers. This is claim (6). □

7. Positive versus resonant: the MC4 dichotomy

Theorem 6.1 upgrades the MC4 completion closure from a conjecture to a theorem with sharp hypotheses. The conditions split the completion landscape into two regimes.

7.1. Positive towers (MC4$^+$).

A strong completion tower is *positive* if the filtration comes from a genuine $\mathbb{Z}_{\geq 1}$-grading preserved by all A_∞-operations, so axiom (4) holds with equality.

Corollary 7.1 (MC4$^+$ is closed)**.** *For any positive strong completion tower $\mathcal{A}$, completed bar-cobar duality holds unconditionally: the counit $\widehat{\epsilon}$ and the unit $\widehat{\eta}$ are quasi-isomorphisms.*

Proof. Positive towers are strong completion towers, so Theorem 6.1 applies. □

This closes the MC4$^+$ problem for all positive towers, including $\mathcal{W}_{1+\infty}$, affine Yangians with positive filtration, and positive RTT towers.

7.2. Resonant towers (MC4^0).

A strong completion tower is *resonant* if there exists a finite-dimensional weight-zero piece $R_{\mathcal{A}} \subset \mathcal{A}$ on which higher operations can preserve filtration degree. The *resonance rank* is $\rho(\mathcal{A}) := \dim H^*(R_{\mathcal{A}}, m_1^{R_{\mathcal{A}}})$. For $\rho = 0$ the tower is positive and Corollary 7.1 applies. For $0 < \rho < \infty$ (the *finite resonance regime*) the strong filtration axiom still holds on the ambient tower and Theorem 6.1 produces a completed bar-cobar quasi-isomorphism with extra cohomology concentrated in the resonance sector. For $\rho = \infty$ the tower is wild and lies outside the present framework.

Theorem 7.2 (W_N rigidity)**.** *Let $\mathcal{A} = W_N$ be the principal finite-type $\mathcal{W}$-algebra with generators of conformal weights $2, 3, \ldots, N$. The standard conformal-weight filtration makes W_N into a strong completion tower in the finite resonance regime. For every cohomological degree m and every conformal weight $w \geq 0$, the transition map*

$$H^m(\overline{B}^{\mathrm{ch}}(W_{N+1}))_w \xrightarrow{\sim} H^m(\overline{B}^{\mathrm{ch}}(W_N))_w$$

is an isomorphism for all $N \geq w$. Consequently the tower

$$\mathcal{W}_\infty := \varprojlim_N W_N$$

admits a completed bar-cobar duality: the counit $\widehat{\epsilon}_{\mathcal{W}_\infty}$ is a quasi-isomorphism.

Proof. W_N satisfies the hypotheses of Proposition 3.3(c) at each finite stage. The transition map $W_{N+1} \to W_N$ forgets the generator of weight $N+1$, which is a strict morphism of curved chiral A_∞-algebras. A bar chain of total conformal weight w can involve only generators of weight $\leq w$; once $N \geq w$, all such generators already occur in W_N, so the weight-w subcomplex is identical in every later stage.

Proposition 5.1 then yields the Mittag-Leffler property on each weight slice, and Theorem 6.1 gives completed bar-cobar duality for the limit $\mathcal{W}_\infty$. The finite resonance rank is computed from the scalar part of the W-OPEs at the weight-zero resonance sector and is finite at every stage. □

Remark 7.3 (Twenty-one conjectures resolved). Theorem 7.2 resolves twenty-one standing conjectures in the completion frontier for principal $\mathcal{W}$-algebras, catalogued in the monograph as the W_N rigidity conjectures for $N = 2, \ldots, 8$ together with the asymptotic stabilization, pro-nilpotence, surjectivity, and uniqueness conjectures. Each of these conjectures asked for a specific completion property; Theorem 7.2 produces them all as consequences of the strong filtration axiom.

8. Examples

8.1. **$\beta\gamma$ and affine Kac-Moody.** Under the fermion-number filtration the $\beta\gamma$-system is a positive strong completion tower ($\rho = 0$): the single-pole OPE $\beta(z)\gamma(w) \sim 1/(z-w)$ gives a Heisenberg-type contribution at bar degree 2 and no higher-degree operations. Under the conformal grading, the strong filtration axiom fails; the fermion-number refinement is essential.

For $V_k(\mathfrak{g})$ at non-critical level the conformal-weight filtration makes $V_k(\mathfrak{g})$ into a positive strong completion tower: the $J^a \cdot J^b$ OPE has poles of order at most 2, so the binary bar operation has weight shift ≤ 1 and Corollary 7.1 applies directly. At the critical level $k = -h^\vee$ the weight-zero resonance becomes infinite and the framework does not apply.

8.2. **Virasoro, $\mathcal{W}_3$, and the principal series.** Virasoro at central charge c has a single weight-2 generator L with a quartic pole in $T \cdot T$:

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}.$$

The conformal-weight filtration makes Vir_c into a strong completion tower with weight-zero resonance given by the central term $c/2$. The resonance rank is $\rho = 1$, so Vir_c lies in the finite resonance regime and Theorem 6.1 gives a completed bar-cobar quasi-isomorphism.

The $\mathcal{W}_3$ algebra has generators T (weight 2) and W (weight 3); the $W \cdot W$ OPE has a sixth-order pole, hence higher-degree bar operations than Virasoro. The strong filtration axiom still holds; the resonance rank is $\rho = 2$. The same argument handles the entire principal series $\mathcal{W}^k(\mathfrak{g}, f_{\mathrm{prin}})$, and Theorem 7.2 gives rigidity at the level of the tower $\mathcal{W}_\infty = \varprojlim_N W_N$.

8.3. **What is excluded.** The strong filtration axiom fails for the $\beta\gamma$-system under the raw conformal grading, for affine $\widehat{\mathfrak{g}}$ at critical level, for algebras with non-additive spectral parameters, and for the open sector of Swiss-cheese algebras where operations act across two colors. A different filtration can sometimes restore axiom (4), but the raw conformal grading alone is insufficient.

8.4. **Summary table.**

Algebra	Filtration	Strong?	ρ	Regime
$V_k(\mathfrak{g})$, $k \neq -h^\vee$	conformal	yes	0	$\mathrm{MC4}^+$
V_Λ	conformal	yes	0	$\mathrm{MC4}^+$
$\beta\gamma$	fermion number	yes	0	$\mathrm{MC4}^+$
$\beta\gamma$	conformal	no	n/a	n/a
Vir_c	conformal	yes	1	$\mathrm{MC4}^0$
$\mathcal{W}_3$	conformal	yes	2	$\mathrm{MC4}^0$
W_N	conformal	yes	$< \infty$	$\mathrm{MC4}^0$
$V_{-h^\vee}(\mathfrak{g})$	conformal	yes	∞	wild

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Appendix A. Note added in proof: $\mathrm{MC4}^0$ UNCONDITIONAL and non-principal hook-type extension

The 2026-04-16 closure wave upgraded $\mathrm{MC4}^0$ from a tractable-on-finite-resonance regime to UNCONDITIONAL at generic parameters for class-M principal algebras, and supplied a parabolic-screening extension to the non-principal hook-type sector.

$\mathrm{MC4}^0$ UNCONDITIONAL at generic parameters.

Theorem A.1 ($\mathrm{MC4}^0$ UNCONDITIONAL, class-M principal)**.** *Let $\mathcal{A}$ be a strong completion tower of class-M principal type (Virasoro, principal $\mathcal{W}_N$), at generic parameters (non-resonant central charge, non-critical level). Then the $\mathrm{MC4}^0$ completion problem of §7 admits an UNCONDITIONAL solution via explicit Steenrod–Denis–Restivo (SDR) data:*

(i) ***Virasoro.*** *SDR constructed via the Wakimoto one-step realisation: $L_n = -{:}\beta\partial\gamma{:}_n - \alpha_0(n+1)\beta_n$ (free-field Coulomb-gas), with homotopy $h_{\mathrm{htpy}} = (1-\iota p)/(L_0 - h - N + 1)$ invertible at generic c.*

(ii) ***Principal $\mathcal{W}_N$.*** *SDR via Feigin–Frenkel screening operators Q_{α_i}: $\mathcal{W}_N$ realised as $\bigcap_i \ker Q_{\alpha_i}$ inside rank-$(N-1)$ Heisenberg, with the same invertible homotopy h_{htpy} at generic Ψ.*

Inscribed as the $\mathrm{MC4}^0$ upgrade in Vol I `chapters/theory/bar_cobar_adjunction_curved.tex`*, referenced at* `thm:completed-bar-cobar-strong` *as part of the 2026-04-16 wave closure.*

Proof sketch. The homotopy $h_{\mathrm{htpy}} = (1-\iota p)/(L_0 - h - N + 1)$ is invertible on $\mathcal{F}^{\geq 1}$ (the positive-weight piece) once the denominator $L_0 - h - N + 1$ has no zero eigenvalue on the relevant weight stratum; at generic c (respectively generic Ψ) the Fock-module / Feigin–Frenkel screening kernel is irreducible at every weight, so the denominator is invertible at every stage of the completion tower. The SDR data then assembles the completed MC element without resort to finite resonance rank: $\rho(\mathcal{A})$ is bounded but its finiteness is no longer a hypothesis of the theorem, only a structural feature of generic-parameter class-M principal algebras. □

Non-principal hook-type extension (parabolic screenings).

Theorem A.2 (Non-principal hook-type, $r \leq N-3$)**.** *For non-principal $\mathcal{W}$-algebras $\mathcal{W}^k(\mathfrak{sl}_N, f)$ of hook-type with partition height $r \leq N-3$, the MC4^0 completion admits an UNCONDITIONAL SDR via parabolic-screening operators, in the Kac–Roan–Wakimoto 2003 and Arakawa 2007 framework:*

$$\mathcal{W}^k(\mathfrak{sl}_N, f_{\mathrm{hook},r}) \;=\; \bigcap_{\alpha \in \Delta_P^+} \ker Q_\alpha \quad \subset \quad V_{k'}(\mathfrak{l}_P)^{\otimes n_P},$$

with P the parabolic corresponding to the hook partition, $\mathfrak{l}_P$ the Levi, and n_P the Heisenberg rank.

Proof sketch. Parabolic screenings extend Feigin–Frenkel to non-principal nilpotents in Young-diagram hook-type. The MC4^0 SDR construction carries through verbatim with parabolic screening kernels replacing the principal Feigin–Frenkel intersection, provided the hook height is bounded $r \leq N-3$ (which guarantees the screening-kernel Koszul-locus preservation). □

Subregular and minimal: reduction to class C SDR. Subregular $\mathcal{W}$-algebras (e.g. Bershadsky–Polyakov BP_k) and minimal $\mathcal{W}$-algebras reduce to the class C $\beta\gamma$ SDR via the principal $\mathcal{W}_{N-2} \times \mathfrak{sl}_2 \times \beta\gamma$ decomposition (subregular) and the coset decomposition (minimal). The Arakawa convention is used throughout: $c(\mathrm{BP}_k) = 2 - 24(k+1)^2/(k+3)$ and $K_{\mathrm{BP}}(k) := c(k) + c(-k-6) \equiv 196$ in $\mathbb{Q}(k)$ (inscribed in Vol I standalone `bp_self_duality.tex` as `thm:bp-koszul-conductor-polynomial`). Fateev–Lukyanov convention gives $K_{\mathrm{BP}}^{\mathrm{FL}}(k) = -12(k+3) - 48/(k+3)$, meromorphic with a pole at $k=-3$; the two conventions describe the same algebra via level reparametrisation.

Summary table update. The row for Vir_c and $\mathcal{W}_N$ is UPGRADED to "MC4^0 UNCONDITIONAL at generic parameters":

Algebra	Filtration	Strong?	Generic status	Regime
$V_k(\mathfrak{g})$, $k \neq -h^\vee$	conformal	yes	unconditional	MC4$^+$
V_Λ	conformal	yes	unconditional	MC4$^+$
$\beta\gamma$ (fermion number)	fermion	yes	unconditional	MC4$^+$
Vir_c, c generic	conformal	yes	UNCONDITIONAL	MC4^0 (Wakimoto SD
$\mathcal{W}_N$, Ψ generic	conformal	yes	UNCONDITIONAL	MC4^0 (FF screening
$\mathcal{W}(\mathfrak{sl}_N, f_{\mathrm{hook},r\leq N-3})$	conformal	yes	unconditional	parabolic screening
BP_k, subregular	conformal	yes	unconditional	reduction to $\beta\gamma$
$V_{-h^\vee}(\mathfrak{g})$	conformal	yes	excluded (Feigin–Frenkel center)	wild

The only residual conjectural content of MC4 is at subregular or minimal $\mathcal{W}$-algebras not reducible to the class-C SDR and at $V_{-h^\vee}(\mathfrak{g})$ (critical level), where the Feigin–Frenkel centre makes $\rho = \infty$. These are genuine remaining mathematics, not completion-machinery plumbing.

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